

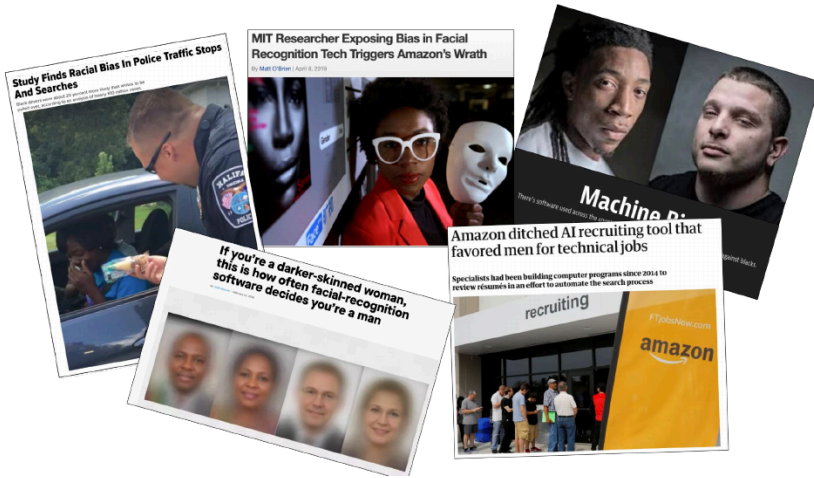
# 2 Algorithmic Fairness

# Algorithmic discrimination and its harms

**Allocation:** access to jobs, credit, education, or other opportunities.

**Quality of service:** how well a system works for different people.

**Representation:** reinforcing stereotypes, demeaning people, or making groups invisible.



Barocas, Hardt, and Narayanan (2023), Chapter 1.

# Scope

**Algorithmic fairness:** computational methods for assessing and mitigating algorithmic discrimination.

These methods can work alongside:

- Investigative reporting and independent audits
- Institutional policy and legal protections
- Advocacy and community organizing

The response may include changing the institution's process or deciding against deployment.

Abebe et al. ([2020](#)).

# Automated decision-making

**Automated decision-making (ADM):** using technology to assist or replace human judgment in a decision.

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Applicant records → Model score → Staff review → Award decision

---

A score or recommendation can affect a decision even when a person makes the final choice.

ADM can use fixed rules, statistical models, or machine learning.

# Anti-discrimination protections

## Ontario Human Rights Code

| Provision  | Protection                                                                            |
|------------|---------------------------------------------------------------------------------------|
| Section 1  | Equal treatment in services, including education                                      |
| Section 5  | Equal treatment in employment                                                         |
| Section 11 | Protection against adverse-effect discrimination from apparently neutral requirements |

These protections concern discrimination on grounds such as race, sex, disability, age, and creed.

A common rule can disadvantage a protected group. Removing the group attribute does not resolve that concern.

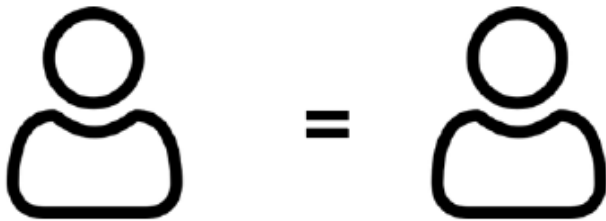
Ontario Human Rights Commission ([2018](#)); Ontario Human Rights Commission ([n.d.](#)); Ontario Human Rights Commission and Law Commission of Ontario ([n.d.](#)), §2C and Part A §3.

# Individual and group fairness

Algorithmic fairness asks when it is acceptable for **automated decision systems (ADS)** to treat people differently.

Motivations include **legal compliance, company policy, and ethics.**

## Individual fairness



Similar applicants should receive similar treatment.

The task-specific similarity comparison needs justification.

## Group fairness



Compare a chosen statistic across groups, such as award rates.

Equal rates alone do not ensure similar treatment of particular applicants.

Dwork et al. ([2012](#)); Barocas, Hardt, and Narayanan ([2023](#)), Chapter 3.

# Fairness through unawareness?

**Baseline:** remove group membership from the model's inputs, then train and apply the model as usual.

Use the same prediction rule for everyone.

**Why it seems reasonable:** the model cannot use the excluded attribute directly.

**What it does not ensure:** other inputs, training labels, and resulting decisions may still differ systematically across groups.

Barocas, Hardt, and Narayanan (2023), Chapter 3, "Groups in the population", pp. 9–11.

# Other inputs can encode group membership

**Example:** the rule awards scholarships to applicants from school P.

Group membership is not an input; school attended is.

|                   | Group <i>a</i> | Group <i>b</i> |
|-------------------|----------------|----------------|
| Applicants        | 100            | 100            |
| Attended school P | 80             | 20             |

The same rule awards scholarships to **80% of group *a*** and **20% of group *b***.

Removing the group label did not remove the association with it.

Barocas, Hardt, and Narayanan (2023), Chapter 3, “Groups in the population”.



# Scholarship prediction



Applicant records  $X$

Predicted probability  $R$

Award decision  $\hat{Y}$

| Variable | Meaning                                          |
|----------|--------------------------------------------------|
| $Y$      | 1 if next-year average $\geq 80\%$ ; 0 otherwise |
| $A$      | Group membership, also retained for evaluation   |

**Proposed policy:** award when the score exceeds a threshold.

Predicting grades does not determine who should receive a scholarship.

# Prediction outcomes

Use a separate matrix for each group.

| Scholarship decision       | Average $\geq 80\%$ ( $Y = 1$ ) | Average $< 80\%$ ( $Y = 0$ ) |
|----------------------------|---------------------------------|------------------------------|
| Award ( $\hat{Y} = 1$ )    | True positive (TP)              | False positive (FP)          |
| No award ( $\hat{Y} = 0$ ) | False negative (FN)             | True negative (TN)           |

$$\text{True positive rate} = \frac{\text{TP}}{\text{TP} + \text{FN}}, \quad \text{False positive rate} = \frac{\text{FP}}{\text{FP} + \text{TN}}.$$

Barocas, Hardt, and Narayanan (2023), Chapter 3, pp. 13–15.

# Statistical fairness notions

## Demographic parity (independence)

$$P(\widehat{Y} = 1 \mid A = a) = P(\widehat{Y} = 1 \mid A = b).$$

Equal award rates across groups.

Calders, Kamiran, and Pechenizkiy ([2009](#)).

# Statistical fairness notions

## Demographic parity (independence)

$$P(\widehat{Y} = 1 \mid A = a) = P(\widehat{Y} = 1 \mid A = b).$$

Equal award rates across groups.

## Equal opportunity

$$P(\widehat{Y} = 1 \mid Y = 1, A = a) = P(\widehat{Y} = 1 \mid Y = 1, A = b).$$

Among those with average  $\geq 80\%$ , equal award rates across groups.

Hardt, Price, and Srebro (2016).

# Statistical fairness notions

## Demographic parity (independence)

$$P(\widehat{Y} = 1 \mid A = a) = P(\widehat{Y} = 1 \mid A = b).$$

Equal award rates across groups.

## Equal opportunity

$$P(\widehat{Y} = 1 \mid Y = 1, A = a) = P(\widehat{Y} = 1 \mid Y = 1, A = b).$$

Among those with average  $\geq 80\%$ , equal award rates across groups.

## Equalized odds (separation)

$$P(\widehat{Y} = 1 \mid Y = y, A = a) = P(\widehat{Y} = 1 \mid Y = y, A = b).$$

Equal award rates within **each** average grade category:  $y = 0$  and  $y = 1$ .

Hardt, Price, and Srebro (2016).

# Group comparisons

In our scholarship example:

| Concern                | Comparison                                      | Statistical criterion |
|------------------------|-------------------------------------------------|-----------------------|
| Access to scholarships | Award rates across groups                       | Independence          |
| Prediction errors      | Award rates within each average grade category  | Separation            |
| What an award predicts | Fraction of recipients with average $\geq 80\%$ | Predictive parity     |

**Which comparison is appropriate depends on what  $Y$  measures and how the decision affects people.**

These are statistical criteria. None is, by itself, a test of compliance with human rights law.

Barocas, Hardt, and Narayanan (2023), Chapters 3–4; Ontario Human Rights Commission and Law Commission of Ontario (n.d.), §2A and Part A §3.

# Sufficiency

$$Y \perp A \mid R$$

For score  $R = r(X) \in \mathbb{R}$ , outcome frequency at a given score is the same across groups:

$$P(Y = 1 \mid R = r, A = a) = P(Y = 1 \mid R = r, A = b).$$

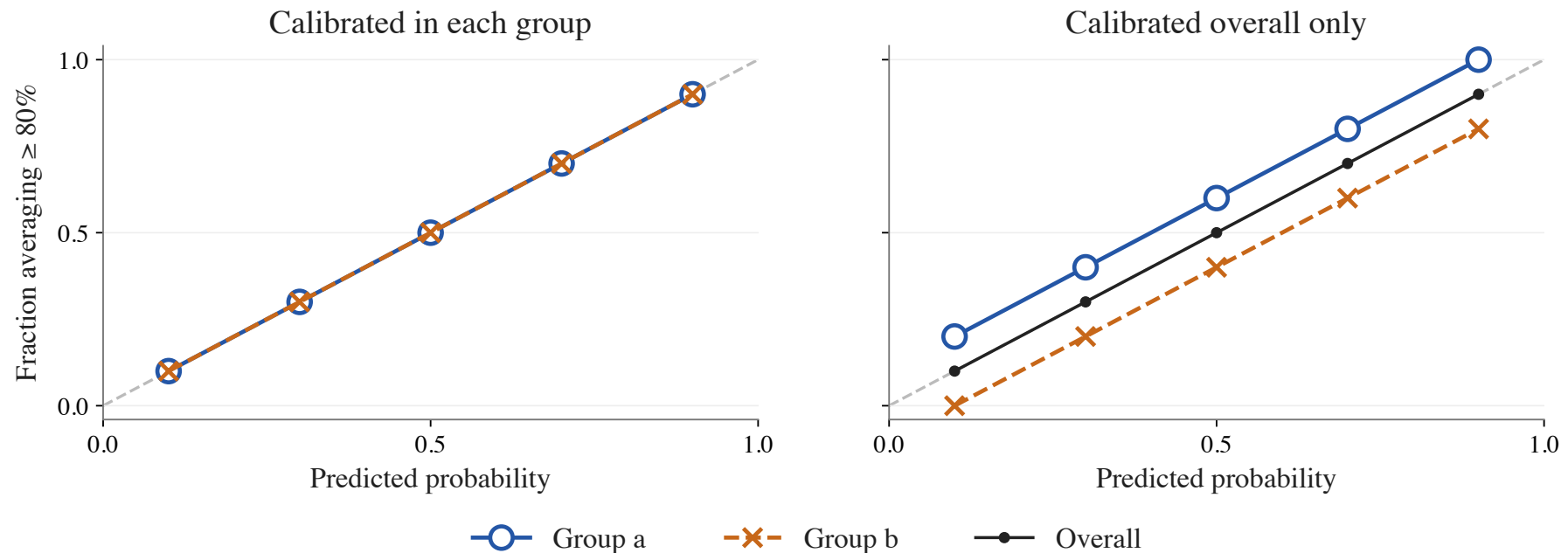
For a **binary prediction**: equal positive predictive value (PPV) **and** negative predictive value (NPV).

*Predictive parity* usually requires equal PPV alone.

Barocas, Hardt, and Narayanan (2023), Chapter 3, pp. 17–18.

# Calibration by group

$$P(Y = 1 \mid R = r, A = a) = r.$$



At score 0.7 (right), the fraction with average  $\geq 80\%$  is 80% in group *a*, 60% in group *b*, 70% overall.

Constructed counts. Barocas, Hardt, and Narayanan (2023), Chapter 3, pp. 18–19.



# Independence and separation

**Base rate:**  $\pi_a = P(Y = 1 \mid A = a)$ , the fraction averaging at least 80%.

**Award rate:**  $q_a = P(\widehat{Y} = 1 \mid A = a)$ .

Split over the two possible outcomes of  $Y$ :

$$\begin{aligned} q_a &= \sum_{y \in \{0,1\}} P(\widehat{Y} = 1, Y = y \mid A = a) \\ &= \sum_{y \in \{0,1\}} P(\widehat{Y} = 1 \mid Y = y, A = a) P(Y = y \mid A = a) \\ &= \text{TPR}_a \pi_a + \text{FPR}_a (1 - \pi_a). \end{aligned}$$

With equalized odds, the common rates are  $u = \text{TPR}$  and  $v = \text{FPR}$ :

$$q_a - q_b = (u - v)(\pi_a - \pi_b).$$

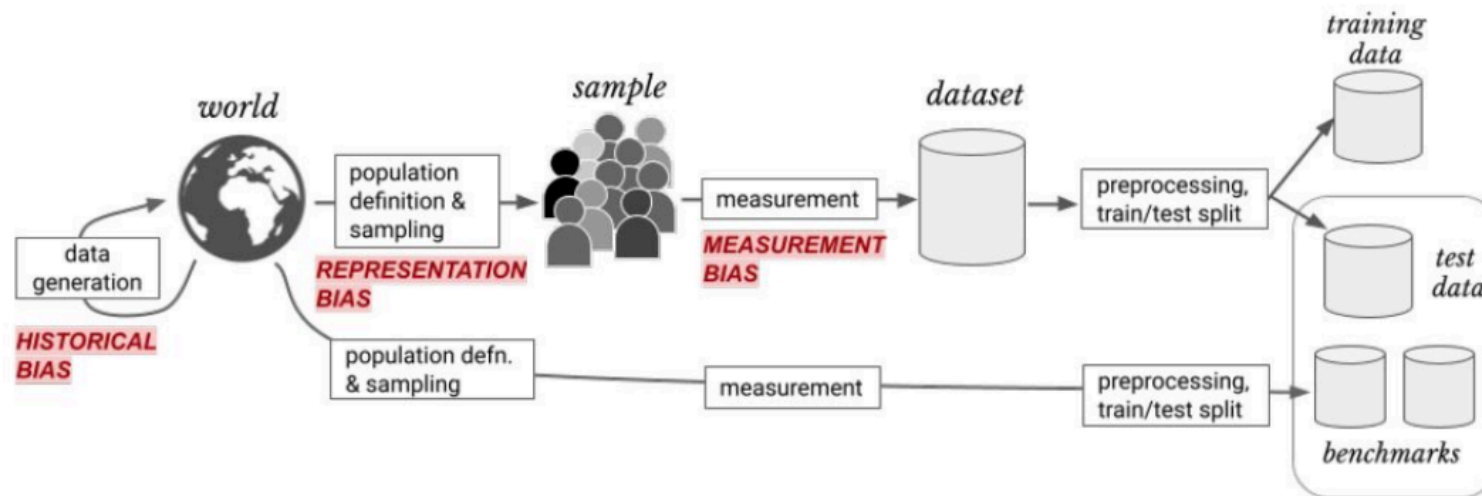
Unequal base rates and  $u \neq v$  prevent equal award rates.

# Limitations of statistical criteria

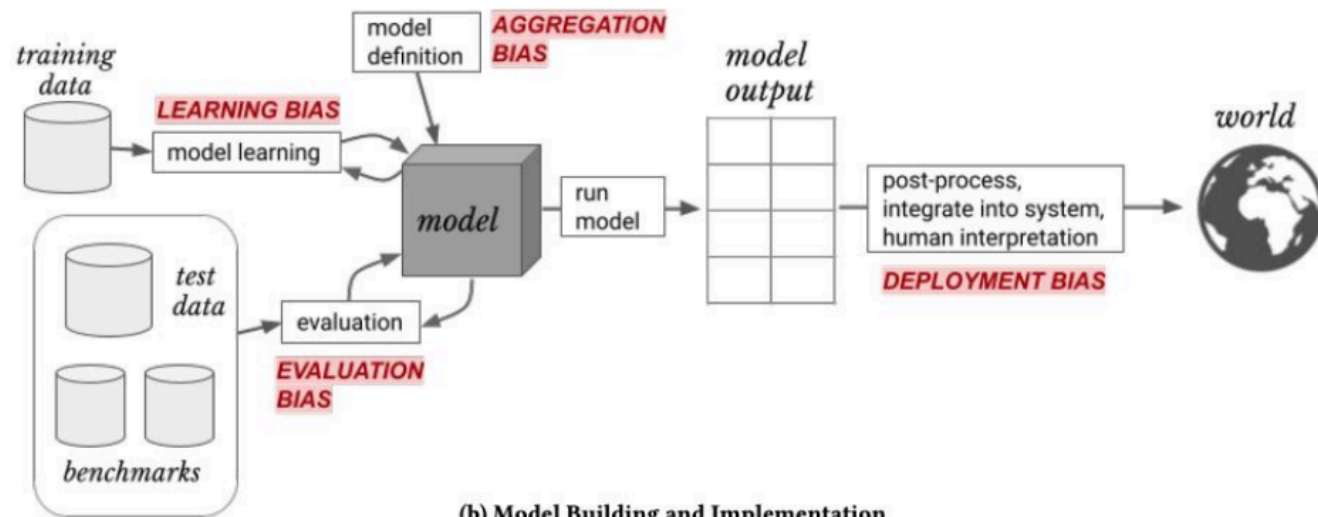
- **Target:** appropriateness and comparability of average grade
- **Population:** whose outcomes are observed or missing
- **Policy:** effects of scholarships on opportunities and outcomes

A fairness assessment also needs evidence about the target, the population, and how decisions are used.

# Sources of harm across the ML lifecycle



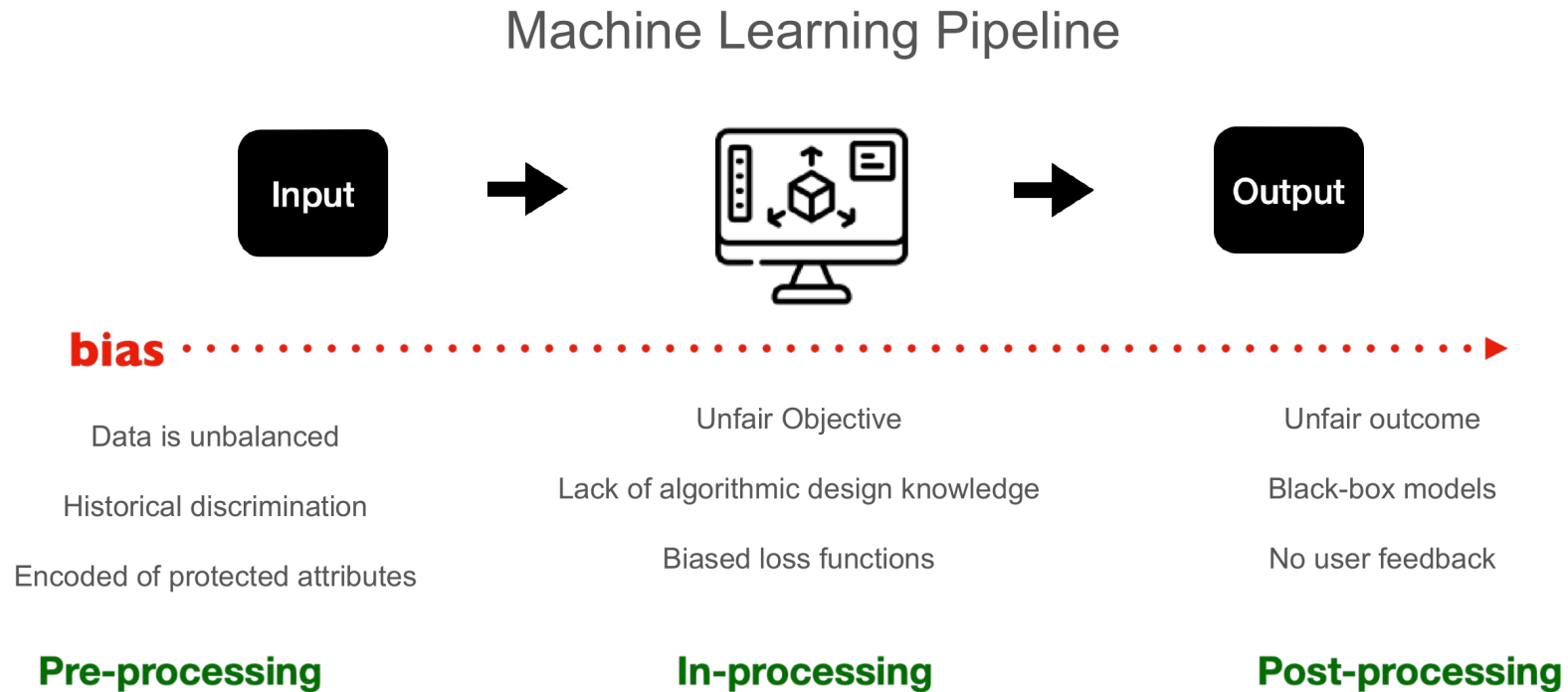
(a) Data Generation



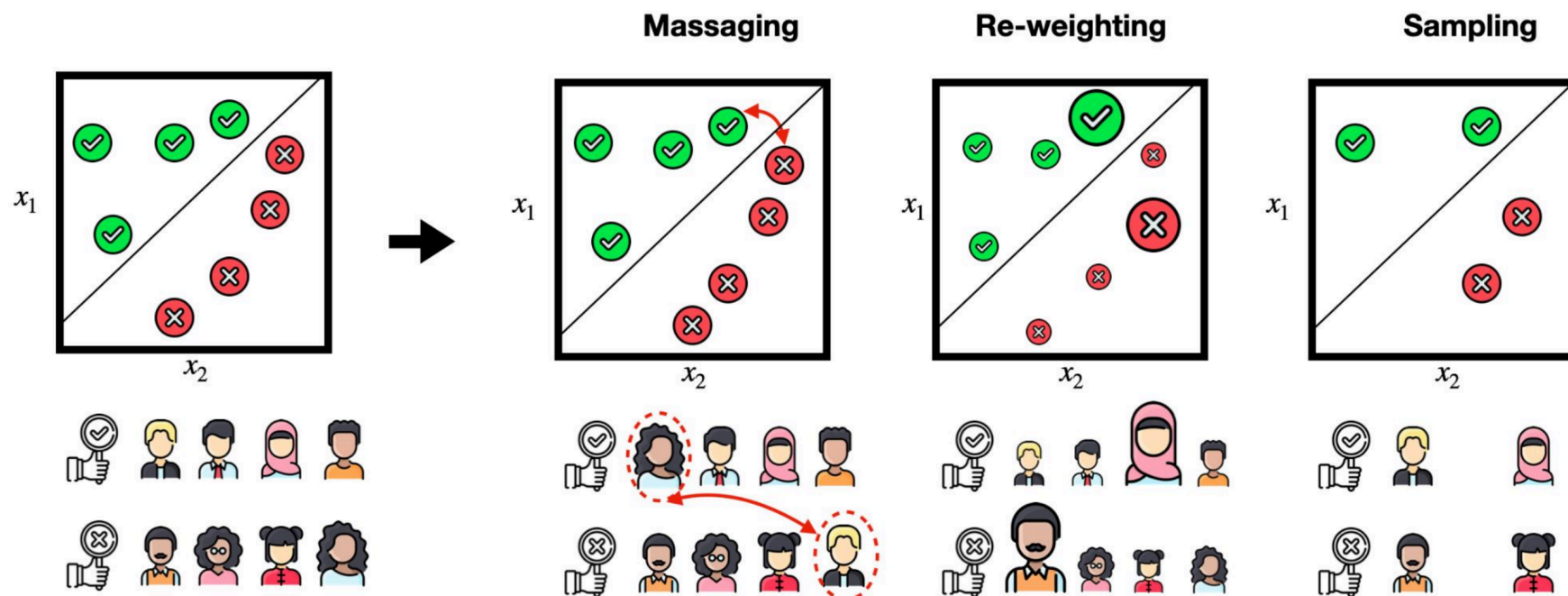
(b) Model Building and Implementation

Suresh and Guttig (2021).

# Fairness interventions



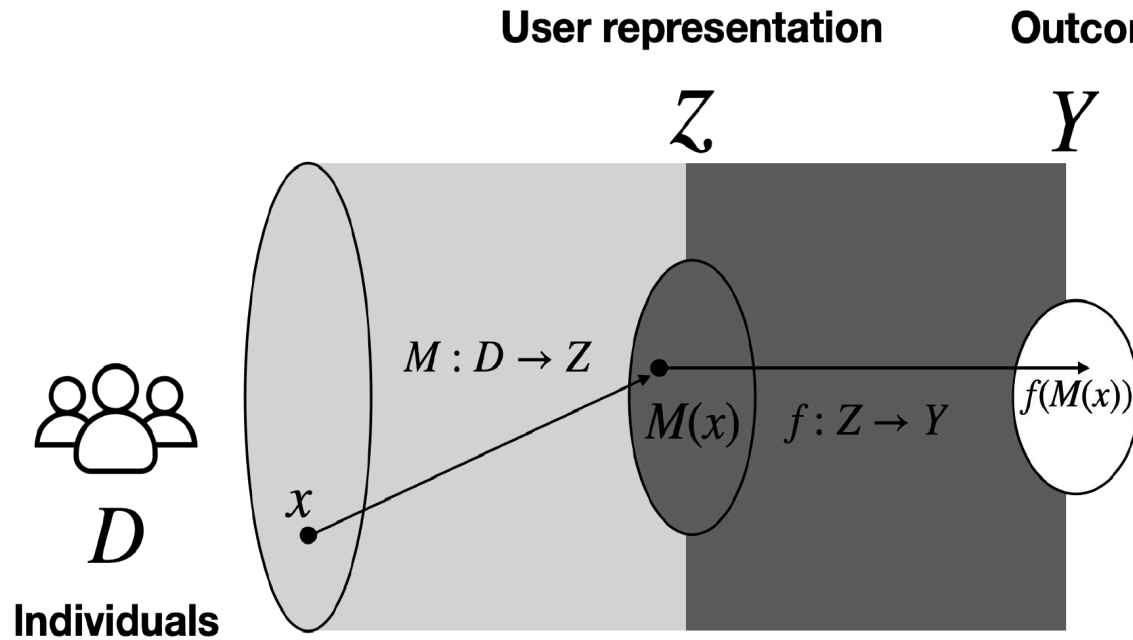
# Data preprocessing



Kamiran, F., & Calders, T. (2012). Data preprocessing techniques for classification without discrimination. *Knowledge and information systems*, 33(1), 1-33.

Alabdulmohsin, I., Schrouff, J., & Koyejo, O. (2022). A Reduction to Binary Approach for Debiasing Multiclass Datasets. *arXiv preprint arXiv:2205.15860*.

# Fair representation learning



**Preserve Performance:**

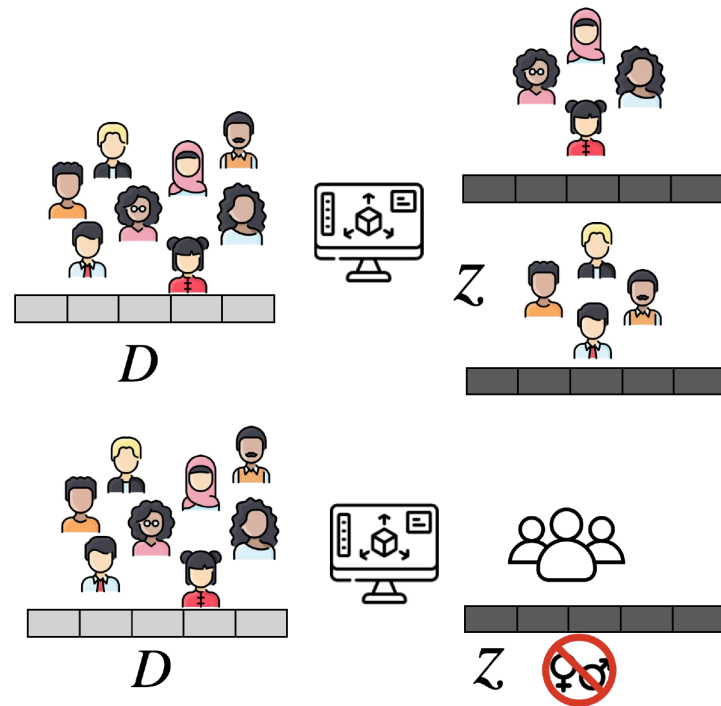
**Reconstruction term:** the learned representation should resemble the original data

**Utility terms:** the learned representation should predict target variable

**+ Fairness**

Zemel, R., Wu, Y., Swersky, K., Pitassi, T., & Dwork, C. (2013, May). Learning fair representations. In International conference on machine learning (pp. 325-333). PMLR

# Fair representation learning: group fairness



Reduce information about group membership in the **whole representation**.

Keep information useful for the prediction task.

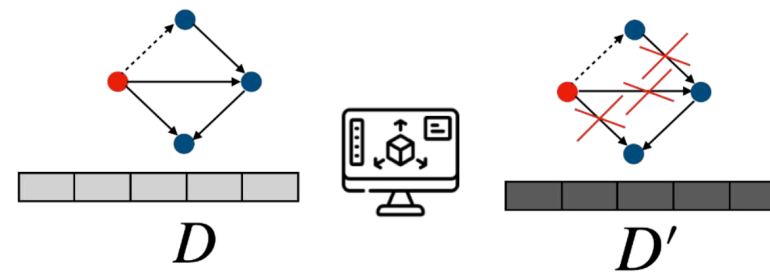
For example, train an adversary to predict group membership, and train the representation to make this difficult.

Louizos et al. (2015); Madras et al. (2018); Locatello et al. (2019)

# Generative models



E.g., Using Generative Adversarial Networks (GANs), Variational Autoencoders, etc.



E.g., Using SCMs (by removing paths from sensitive attributes)

Sattigeri, Prasanna, et al. "Fairness GAN: Generating datasets with fairness properties using a generative adversarial network." *IBM Journal of Research and Development* 63.4/5 (2019): 3-1.

van Breugel, Boris, et al. "Decaf: Generating fair synthetic data using causally-aware generative networks." *Advances in Neural Information Processing Systems* 34 (2021): 22221-22233.



# In-processing techniques

- Supervised learning tasks are often expressed as optimization problems

$$\underset{\theta}{\text{minimize}} \quad f(X, Y; \theta)$$

- The optimization problem: finding the parameters that give the best model w.r.t the desired properties

**Fairness in another desired property of the learned models**

$$g(X, Y; \theta)$$

# In-processing techniques

Fairness as  
Constrained Optimization

$$\begin{array}{ll} \underset{\theta}{\text{minimize}} & f(X, Y; \theta) \\ \text{subject to} & g(X, Y; \theta) \end{array}$$

Fairness as  
Regularizer

$$\underset{\theta}{\text{minimize}} \quad f(X, Y; \theta) + \lambda g(X, Y; \theta)$$

Fairness as  
Multi-objective Optimization

$$\underset{\theta}{\text{minimize}} \quad f(X, Y; \theta) \times g(X, Y; \theta)$$

Choi, Y., Farnadi, G., Babaki, B., & Van den Broeck, G. (2020, April). Learning fair naive bayes classifiers by discovering and eliminating discrimination patterns. In *Proceedings of the AAAI Conference on Artificial Intelligence* (Vol. 34, No. 06, pp. 10077-10084).

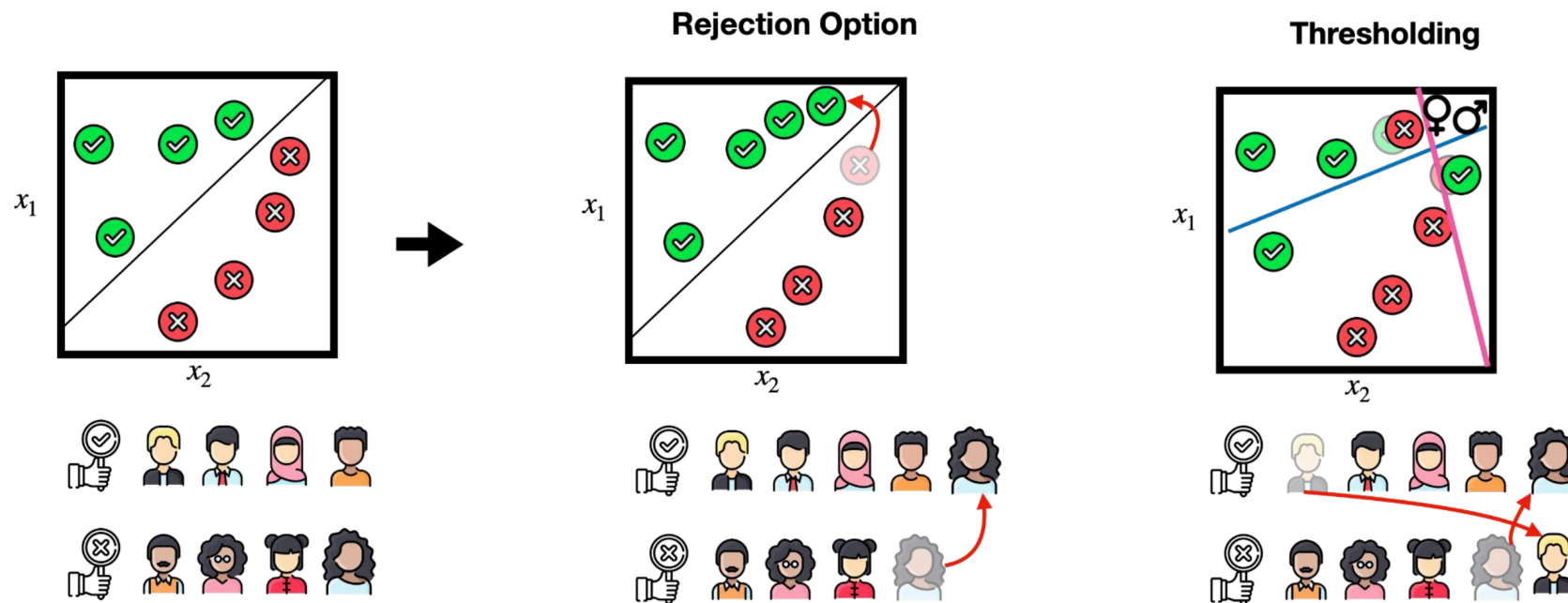
Mohammadi, K., Sivaraman, A., & Farnadi, G. (2022). FETA: Fairness Enforced Verifying, Training, and Predicting Algorithms for Neural Networks. *arXiv preprint arXiv:2206.00553*.

Kamishima, Toshihiro, Shotaro Akaho, and Jun Sakuma. "Fairness-aware learning through regularization approach." 2011 IEEE 11th International Conference on Data Mining Workshops. IEEE, 2011.

A. Agarwal, A. Beygelzimer, M. Dudík, J. Langford, and H. Wallach, "A Reductions Approach to Fair Classification," arXiv.org, 16-Jul-2018. [Online]. Available: <https://arxiv.org/abs/1803.02453>.

Kearns, M., Neel, S., Roth, A., & Wu, Z. S. (2018, July). Preventing fairness gerrymandering: Auditing and learning for subgroup fairness. In *International Conference on Machine Learning* (pp. 2564-2572). PMLR.

# Post-processing techniques



Kamiran, F., Karim, A., & Zhang, X. (2012, December). Decision theory for discrimination-aware classification. In 2012 IEEE 12th International Conference on Data Mining (pp. 924-929). IEEE.

Hardt, M., Price, E. and Srebro, N., 2016. Equality of opportunity in supervised learning. In Advances in neural information processing systems (pp. 3315-3323).

# Trade-offs

|                                                         | Ease of implementation and (re)-use | Scalability | Ease of auditing | Fairness/ Performance tradeoff | Generalization |
|---------------------------------------------------------|-------------------------------------|-------------|------------------|--------------------------------|----------------|
| <b>Pre-processing,</b><br>e.g., representation learning | ✓                                   | ✓           | ✓                |                                | ✓              |
| <b>In-processing,</b><br>e.g., fairness regularizer     |                                     |             | ✓                | ✓                              | ✓              |
| <b>Post-processing,</b><br>e.g., thresholding           |                                     | ✓           | ✓                |                                |                |

An illustrative comparison; properties depend on the method and setting.

# Useful Metrics for A1

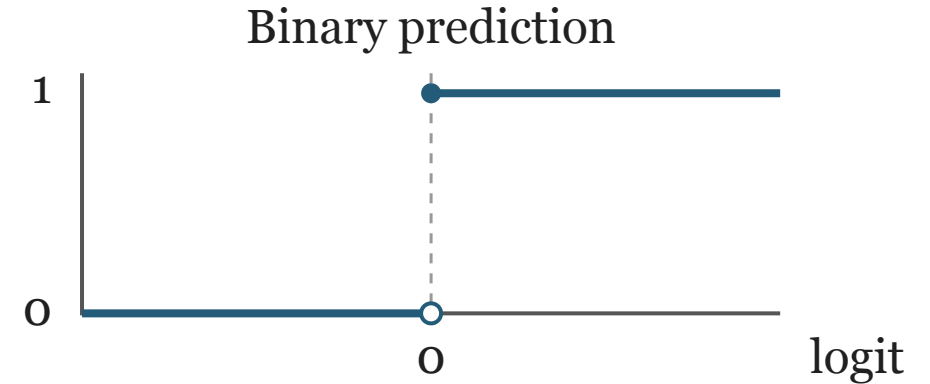
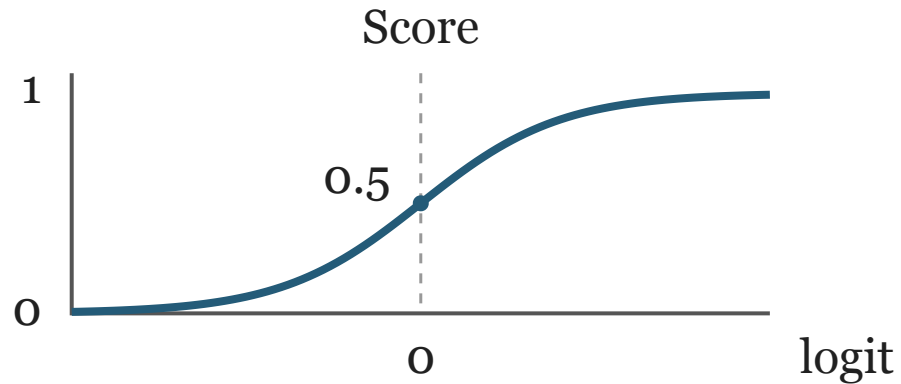
**Adult:**  $Y = 1$  means income above US\$50,000;  $A$  is recorded sex.

| Metric                    | Calculation                                                             |
|---------------------------|-------------------------------------------------------------------------|
| Pearson correlation       | $\rho(X_j, Z) = \frac{\text{Cov}(X_j, Z)}{\text{sd}(X_j) \text{sd}(Z)}$ |
| Accuracy                  | $P(\widehat{Y} = Y)$                                                    |
| Group-reweighted accuracy | $\frac{1}{2} \sum_{a=0}^1 P(\widehat{Y} = Y \mid A = a)$                |
| Demographic-parity gap    | $ P(\widehat{Y} = 1 \mid A = 0) - P(\widehat{Y} = 1 \mid A = 1) $       |

For correlations,  $Z$  can be the outcome  $Y$ , group  $A$ , or prediction  $\widehat{Y}$ .

Within a group, recompute the means and standard deviations using that group.

# Differentiable fairness penalties



The binary decision has zero gradient away from the threshold and is discontinuous at the threshold.

# Individual fairness

1 · Present

## Fairness Through Awareness

Cynthia Dwork\*

Moritz Hardt<sup>†</sup>

Toniann Pitassi<sup>‡</sup>

Omer Reingold<sup>§</sup>

Richard Zemel<sup>¶</sup>

Individual fairness compares treatment using a task-specific similarity metric.

ITCS 2012. Dwork et al. ([2012](#)).

# Similar applicants

1 · Present

Suppose each row contains two applicants deemed similar for this task.

| Similar pair            | Group <i>a</i> : award? | Group <i>b</i> : award? |
|-------------------------|-------------------------|-------------------------|
| Pair 1                  | 1                       | 0                       |
| Pair 2                  | 0                       | 1                       |
| <b>Group award rate</b> | <b>50%</b>              | <b>50%</b>              |

Equal group rates can coexist with different treatment of similar people.

Dwork et al. (2012), §§1–2.



# Distributions over outcomes

1 · Present

$V$ : individuals.  $\mathcal{O}$ : possible decisions.

$$M : V \longrightarrow \Delta(\mathcal{O}).$$

$M(x)$  assigns probabilities to the decisions available for person  $x$ .

**Individual fairness:**

$$\underbrace{D(M(x), M(x'))}_{\text{difference in treatment}} \leq \underbrace{d(x, x')}_{\text{task-specific difference between people}} .$$

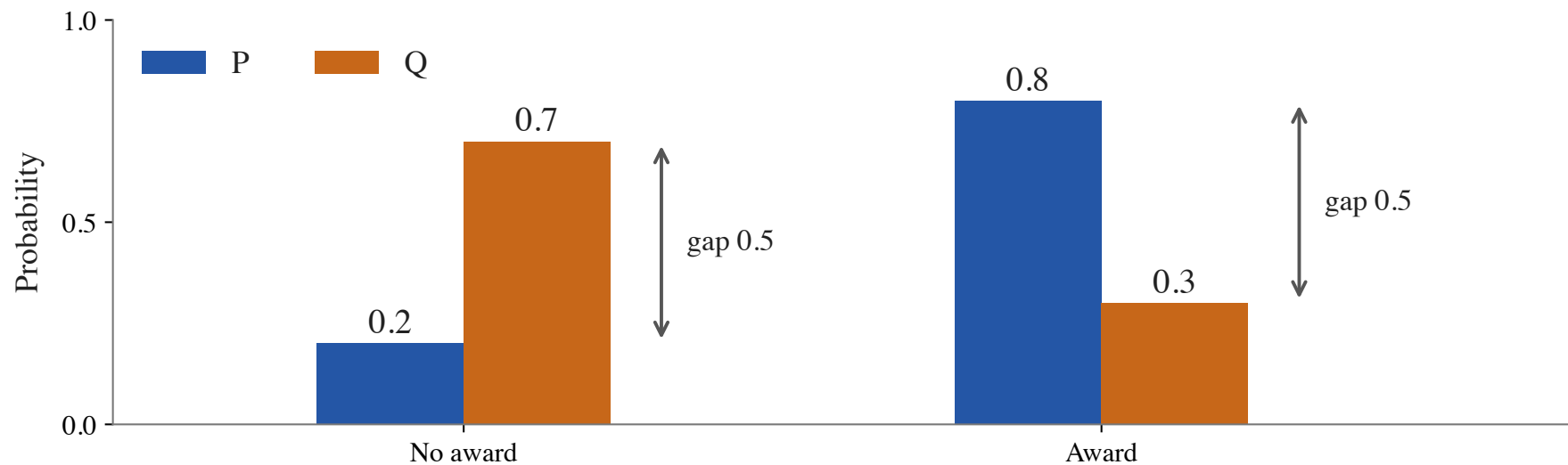
The condition must hold for every pair of individuals.

Dwork et al. (2012), Definition 2.1.

# Total variation distance

1 · Present

$$D_{\text{TV}}(P, Q) = \frac{1}{2} \sum_{o \in \mathcal{O}} |P(o) - Q(o)|.$$



Here  $D_{\text{TV}} = \frac{1}{2} (.5 + .5) = .5$ : the **award-rate gap**.

Example. Dwork et al. (2012), §2.1.

# Binary decisions

1 · Present

Let  $p_x$  be the probability that  $x$  receives a scholarship.

$$M(x) = (1 - p_x, p_x), \quad D_{\text{TV}}(M(x), M(x')) = |p_x - p_{x'}|.$$

For a pair with stipulated distance  $d(x, x') = .10$ :

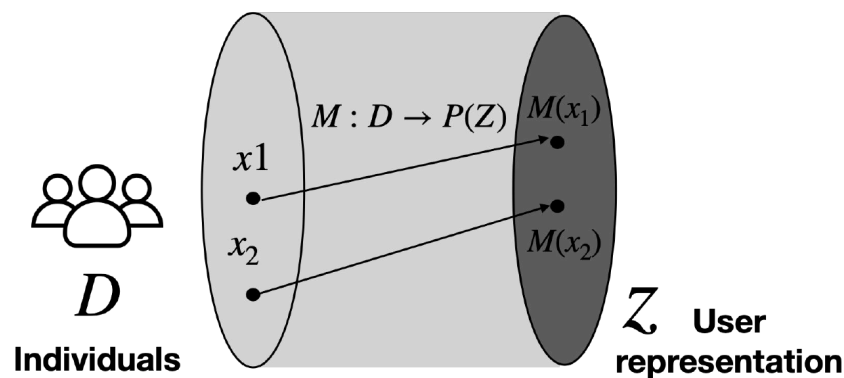
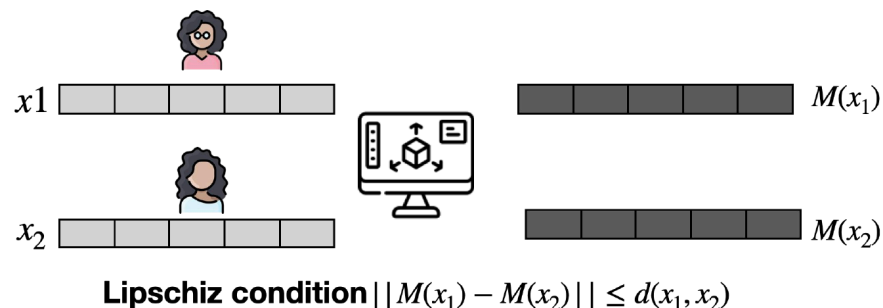
| Rule   | $p_x$ | $p_{x'}$ | Difference | Constraint |
|--------|-------|----------|------------|------------|
| First  | .80   | .30      | .50        | Violated   |
| Second | .55   | .45      | .10        | Satisfied  |

The constraint compares **distributions**, not whether two realized decisions match.

Dwork et al. (2012), §2.1, Eq. (5).

# Fair representation learning: individual fairness

## 1 · Present



## Fairness

- **Similar individuals** should map to **similar distributions**.
- **Task-specific** similarity metric. Ideally captures ground truth or society's best approximation
- Many applications: ranking in recommender systems, financial risk metrics, health metric for treating patients, etc.

Dong, Yushun, et al. "Individual fairness for graph neural networks: A ranking based approach." Proceedings of the 27th ACM SIGKDD Conference on Knowledge Discovery & Data Mining. 2021.

Salganik, Rebecca, Fernando Diaz, and Golnoosh Farnadi. "Analyzing the Effect of Sampling in GNNs on Individual Fairness." *arXiv preprint arXiv:2209.03904* (2022).

# Catalog of evils

## 1 · Present

| Practice                       | Mechanism in Appendix A                                                      |
|--------------------------------|------------------------------------------------------------------------------|
| Explicit discrimination        | Test group membership and assign worse outcomes                              |
| Redundant encoding             | Use other attributes as a substitute for the group label                     |
| Redlining                      | Deny or limit financial services by neighbourhood                            |
| Excluding a population segment | Exclude a segment where the protected group is overrepresented               |
| Self-fulfilling prophecy       | Select unsuitable members, then use their outcomes to discredit the group    |
| Reverse tokenism               | Reject a highly qualified outsider as a token defence against discrimination |

Dwork et al. (2012), Appendix A, “Catalog of Evils”.

# Choosing the similarity metric

1 · Present

The constraint protects pairs that the metric places **close together**.

| Choice                    | Pitfall                                                                       |
|---------------------------|-------------------------------------------------------------------------------|
| Task-relevant similarity  | A convenient feature distance can treat a proxy as a justified difference     |
| Comparisons across groups | Within-group data alone do not justify how people in different groups compare |
| Scale of the distance     | Larger distances permit larger differences in treatment                       |

A metric that separates groups can permit the very practices we want to prevent.

The optimization does not establish that the supplied metric is appropriate.

Dwork et al. (2012), §§2.2 and 6.1.

# Statistical parity

1 · Present

Let  $S, T$  be distributions over people in two groups. Their outcome distributions are

$$\mu_S(o) = \mathbb{E}_{x \sim S}[M(x)(o)], \quad \mu_T(o) = \mathbb{E}_{x \sim T}[M(x)(o)].$$

**Statistical parity (demographic parity):**  $\mu_S = \mu_T$ .

For binary decisions, this is equality of positive-decision rates:

$$P(\widehat{Y} = 1 \mid A = a) = P(\widehat{Y} = 1 \mid A = b).$$

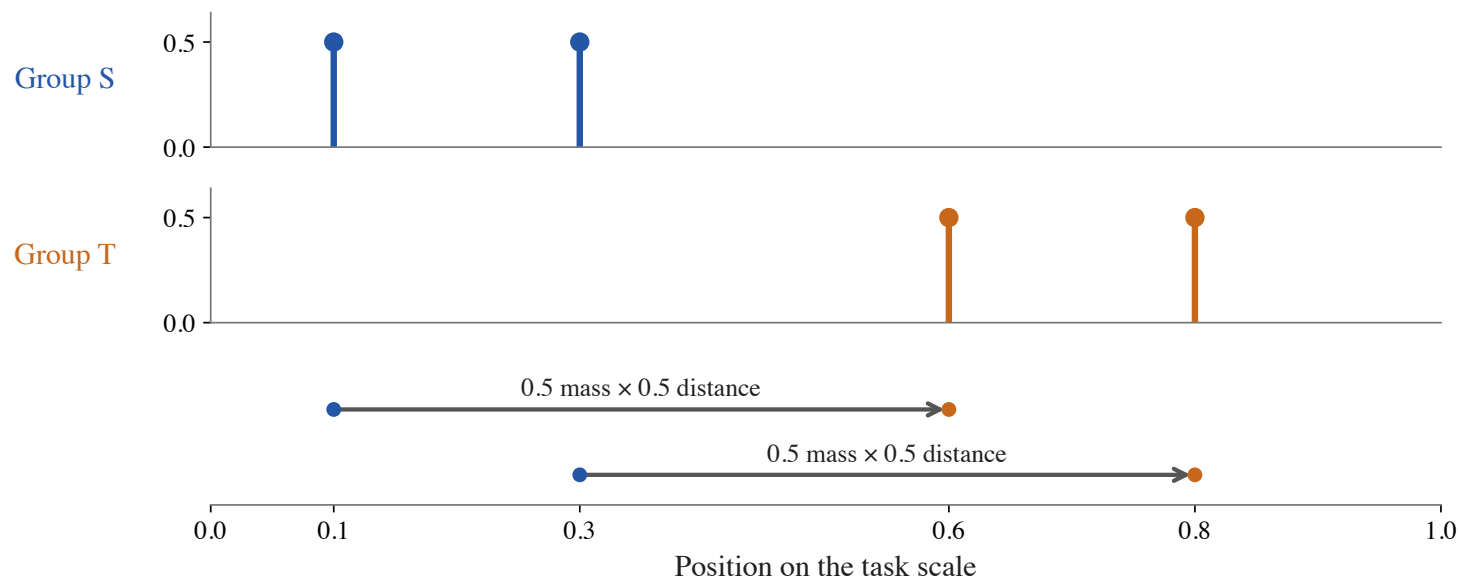
Parity up to  $\varepsilon$  means  $D_{\text{TV}}(\mu_S, \mu_T) \leq \varepsilon$ .

Dwork et al. (2012), §3.

# Earthmover distance

1 · Present

Move probability mass between groups using the task distance  $d$ .



$$d_{\text{EM}}(S, T) = \min_{\gamma \in \Pi(S, T)} \mathbb{E}_{\gamma}[d(x, x')].$$

Here the minimum cost is  $.5 \times .5 + .5 \times .5 = .5$ .

Example. Dwork et al. (2012), Definition 3.3.



# Individual fairness and statistical parity

1 · Present

**Theorem 3.3:** if  $M$  is individually fair using  $D_{\text{TV}}$  and metric  $d$ , then

$$D_{\text{TV}}(\mu_S, \mu_T) \leq d_{\text{EM}}(S, T).$$

Groups close under  $d$  must receive similar outcome distributions.

If all task distances are at most 1, the bound is tight:

$$\max_{M \text{ individually fair}} D_{\text{TV}}(\mu_S, \mu_T) = d_{\text{EM}}(S, T).$$

This is the **largest permitted gap**, not the gap of every fair rule.

Dwork et al. (2012), Lemma 3.1 and Theorem 3.3.

# Two groups on a task scale

1 · Present

Two equally weighted people per group;  $d(x, x') = |x - x'|$ .

| Matched pair | Position in $S$ | Position in $T$ | Distance |
|--------------|-----------------|-----------------|----------|
| 1            | .10             | .60             | .50      |
| 2            | .30             | .80             | .50      |

Minimum average distance:  $d_{\text{EM}}(S, T) = .50$ .

| Individually fair rule | Positive rate in $S$ | In $T$ | Gap |
|------------------------|----------------------|--------|-----|
| $p_x = x$              | .20                  | .70    | .50 |
| $p_x = .50$            | .50                  | .50    | .00 |

Individual fairness **permits** the gap; it does not require it.

Dwork et al. (2012), Theorem 3.3.

# Metric choices

## 2 · Reflect

Choose one discriminatory practice from Appendix A. Which individuals should have a small distance under the similarity metric, and why? 🤔

Answer in class.

**Individual work (30 seconds):** identify a practice and two individuals to compare.

**Pair discussion (90 seconds):** identify the task-relevant evidence that would justify a small distance between these individuals.

# Individual and group fairness

3 · Discuss

How do you interpret the relationship between individual and group fairness? 🤔

Answer in class.

What questions or concerns did the paper raise for you?

# The individual-fairness guarantee

4 · Synthesize

When does individual fairness imply approximate statistical parity, and what still needs justification? 🤔

Answer in class.

Record one precise guarantee and one unresolved choice from the discussion.

# Fair prediction

1 · Present

Fair prediction with disparate impact:  
A study of bias in recidivism prediction instruments

Alexandra Chouldechova \*

Predictive parity and equal error rates can disagree when base rates differ.

Big Data 2017. Chouldechova ([2017](#)).

# COMPAS study

## 1 · Present

Historical COMPAS analysis: Broward County defendants scored in 2013–2014.

| Quantity             | In this paper                     |
|----------------------|-----------------------------------|
| Outcome $Y$          | Recorded two-year recidivism      |
| Group $A$            | Recorded Black / White categories |
| Score $S$            | COMPAS decile, 1–10               |
| Prediction $\hat{Y}$ | High risk: $S > s_{\text{HR}}$    |

The filtered analysis contains **6,150 people**.

Chouldechova ([2017](#)), §1.2.

# Calibration and predictive parity

1 · Present

The paper's **calibration** condition compares groups at the same score:

$$P(Y = 1 \mid S = s, A = a) = P(Y = 1 \mid S = s, A = b).$$

**Predictive parity** compares people above a threshold:

$$P(Y = 1 \mid S > s_{\text{HR}}, A = a) = P(Y = 1 \mid S > s_{\text{HR}}, A = b).$$

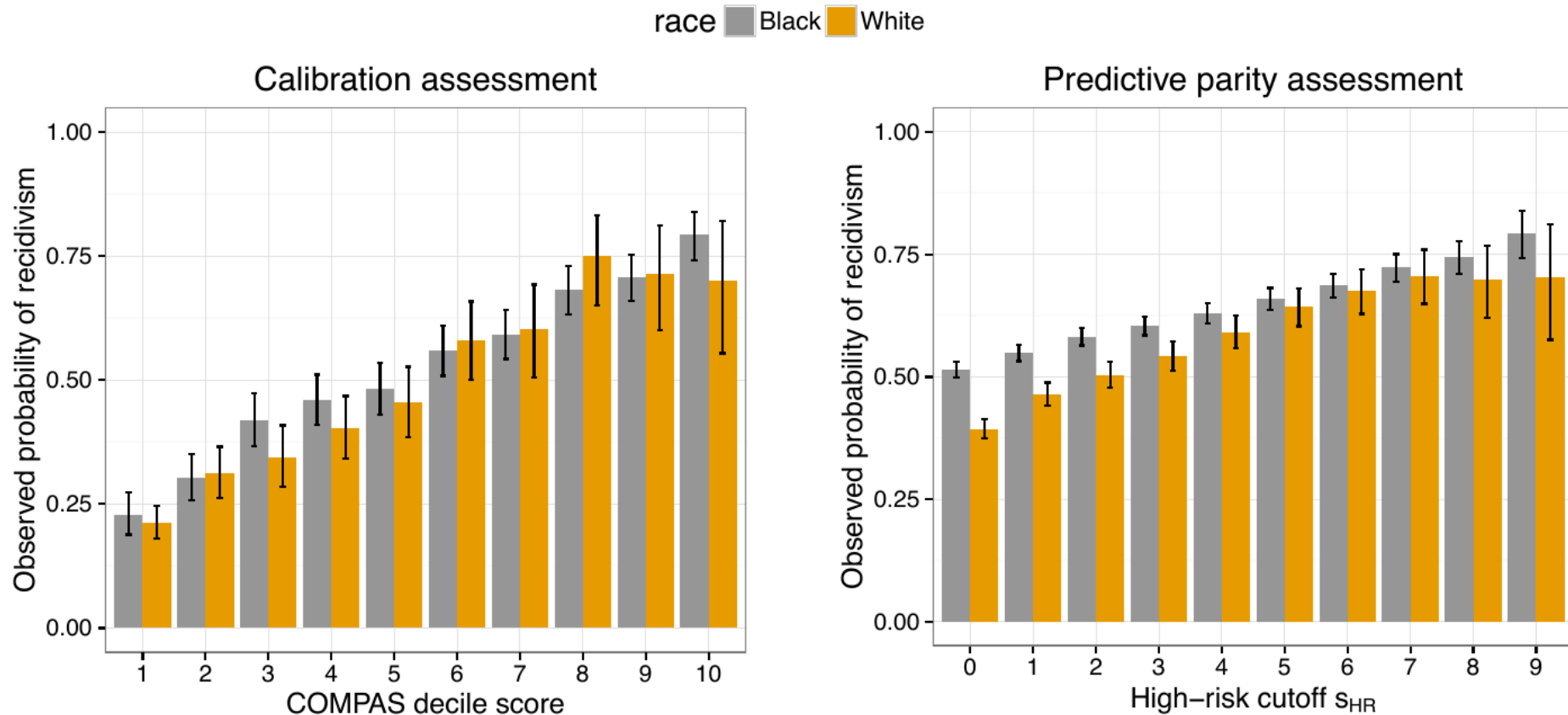
Pooling scores above a threshold can mix them in different proportions across groups.

Chouldechova ([2017](#)), Definitions 1–2.



# Predictive meaning of scores

## 1 · Present



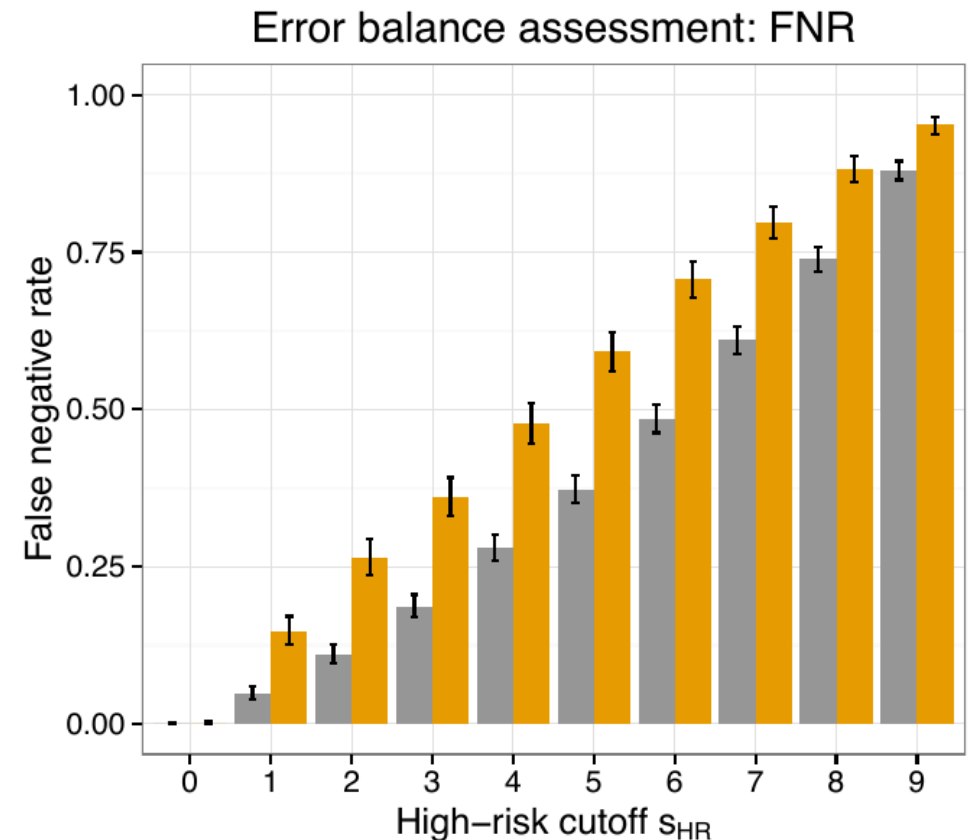
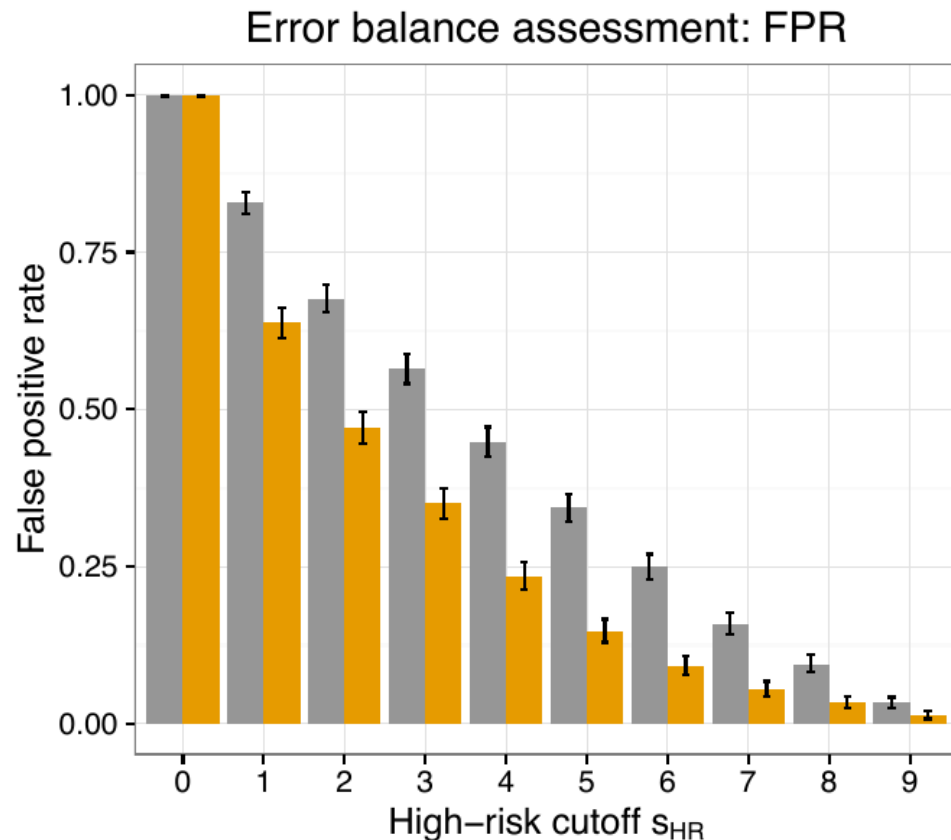
The paper reports approximate agreement, including predictive parity for cutoffs of 4 or higher.

Chouldechova (2017), Fig. 1(a–b). Error bars: 95% confidence intervals.

# Error rates

1 · Present

**Grey: Black defendants. Gold: White defendants.**



At cutoff 4, the group with the higher FPR has the lower FNR.

Chouldechova (2017), Fig. 1(c-d). Error bars: 95% confidence intervals.

# Predictive values and base rates

1 · Present

For group  $a$ , let  $\pi_a = P(Y = 1 \mid A = a)$ .

$$\text{PPV}_a = \frac{\text{TPR}_a \pi_a}{\text{TPR}_a \pi_a + \text{FPR}_a (1 - \pi_a)}.$$

Equivalently,

$$\text{FPR}_a = \frac{\pi_a}{1 - \pi_a} \frac{1 - \text{PPV}_a}{\text{PPV}_a} (1 - \text{FNR}_a).$$

With  $0 < \pi_a < 1$  and common positive TPR and FPR, unequal base rates imply unequal PPV.

Chouldechova ([2017](#)), Eq. (2.6).

# Penalty policy

1 · Present

|                   | $Y = 1$ | $Y = 0$ |
|-------------------|---------|---------|
| $\widehat{Y} = 1$ | TPR     | FPR     |
| $\widehat{Y} = 0$ | FNR     | TNR     |

Rates conditional on column and group.

**Hypothetical policy:** a high-risk label increases the penalty.

$$T = t_{\min} + (t_{\max} - t_{\min})\widehat{Y}.$$

Among people with recorded  $Y = 0$ :

$$\mathbb{E}[T \mid Y = 0, A = a] = t_{\min} + (t_{\max} - t_{\min})\text{FPR}_a.$$

The FPR difference becomes an expected penalty difference, scaled by  $t_{\max} - t_{\min}$ .

Changing how the score is used changes the consequences of its errors.

Chouldechova ([2017](#)), §3, Proposition 3.1 and Corollary 3.1.

# Interpreting the conflict

2 · Reflect

Name one conclusion justified by the identity and one decision it cannot settle. 🤔

Answer in class.

**30 seconds alone:** write the conclusion and its assumptions.

**90 seconds in pairs:** compare the institutional questions left open.

# Choosing a decision rule

3 · Discuss

How should a decision rule be chosen when predictive parity and equal error rates cannot both be satisfied? 🤔

Answer in class.

Consider two uses of a high-risk prediction: **increase the penalty** or **provide additional support**.

How would the consequences of false positives and false negatives differ?

# Incompatibility

4 · Synthesize

Under what assumptions are predictive parity and equal error rates incompatible? What does this result tell us about choosing a decision rule? 🤔

Answer in class.

Explain which conclusions follow from the result and which require additional assumptions or policy decisions.

# Fairness and abstraction

1 · Present

## Fairness and Abstraction in Sociotechnical Systems

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Evaluating fairness requires examining how a model is developed and used, including the decisions made by people and institutions.

FAT\* 2019. Selbst et al. (2019).



# Sociotechnical systems

## 1 · Present

A **sociotechnical system** includes technical components, people, institutions, and the interactions among them.

### Technical components

Data, models, software, interfaces

### Social components

People, roles, rules, incentives, institutions

People determine how a model is built and used. The model can also change how people work and make decisions.

For example, a risk score may inform a judge's decision; court procedures determine how the score is used and challenged.

Selbst et al. (2019), §§2.1 and 3.

# Framing trap

1 · Present

**Defining the system too narrowly can exclude decisions that affect fairness.**

A model produces predictions; people and institutions determine how those predictions are used.

A fairness guarantee about model outputs may not apply to final decisions.

Selbst et al. ([2019](#)), §2.1.

# Portability trap

1 · Present

**A method developed for one social context may be inappropriate in another.**

Institutional goals, decision procedures, and fairness concerns can differ across settings.

These differences are not fully described by changes in the distribution of features and labels.

Selbst et al. ([2019](#)), §2.2.

# Formalism trap

1 · Present

**A mathematical criterion captures only part of what fairness can mean.**

Fairness can depend on the decision process, the social context, and disagreement about acceptable treatment.

Mathematics alone cannot determine which criterion is appropriate, or whether any proposed criterion is sufficient.

Selbst et al. ([2019](#)), §2.3.

# Ripple effect trap

1 · Present

**Introducing a model can change the system in which it operates.**

People may change their behaviour, responsibilities, or reliance on other sources of evidence.

Institutions may give greater weight to objectives that the model measures.

Selbst et al. ([2019](#)), §2.4.

# Solutionism trap

1 · Present

**The appropriate response to a problem may not involve a technical system.**

Starting with a model can exclude other interventions from consideration.

Compare the proposed system with changes to policy or practice, including the option not to build it.

Selbst et al. ([2019](#)), §2.5.

# Questions about the traps

2 · Reflect

Which trap raises a question or objection for you? 🤔

Answer in class.

**Individual work (30 seconds):** note a question, objection, or example.

**Pair discussion (90 seconds):** compare your responses.

# Discussion

3 · Discuss

What do you find convincing or unconvincing in the paper's argument?



Answer in class.

What would you like to discuss further?



# Implications for research and practice

## 4 · Synthesize

What, if anything, would you change in your research or practice after reading this paper? 🤔

Answer in class.

Explain your reasons. You may also identify a question that remains unresolved.

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